

Physical Chemistry (I)

Lecture 22

Chemical Reactions

Electrochemistry



In the Last Class

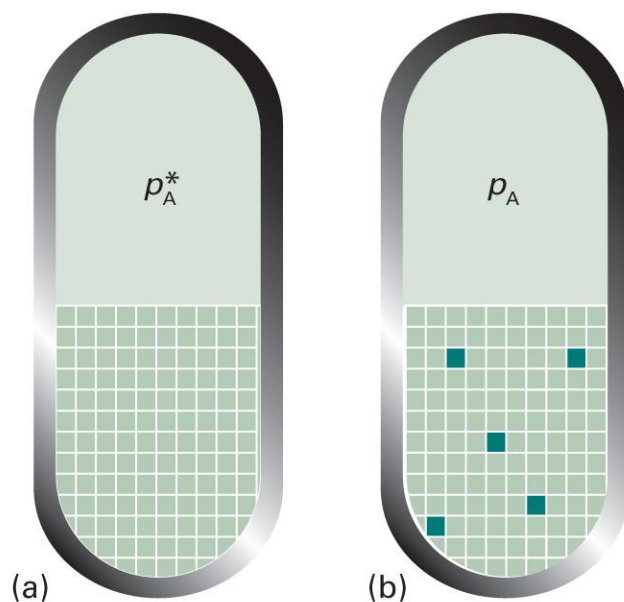
- Colligative properties
 - Lowering of vapor pressure
 - Elevation of boiling point
 - Depression of freezing point
 - Osmotic pressure
- Chemical reactions
 - Reaction quotients and equilibrium constants



Last class

Origin of Colligative Properties

- Due to the reduction of the chemical potential of the liquid solvent as a result of the presence of solute

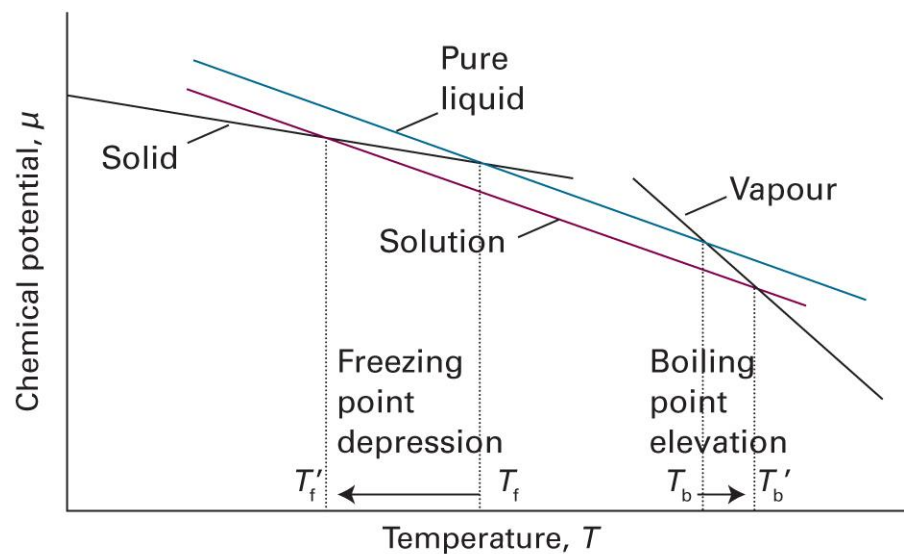


(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.7)

Last class

Changes in Phase Transitions

$$\mu_A = \mu_A^* + RT \ln x_A \quad (x_A < 1)$$



Virial Expansion

For an ideal solution (now solute = J),

$$\Pi = [J]RT$$

The osmotic virial expansion is

$$\Pi = [J]RT(1 + B[J] + C[J]^2 + \dots)$$

where $B, C, \dots$ are the **osmotic virial coefficients**.

Case of Ionic Solutes



- van't Hoff factor i

$$\Delta T_b = iK_b b$$

$$\Delta T_f = iK_f b$$

$$\Pi = i[B]RT$$

Last class

Extent of Reaction ξ



Species	Initial	→	Change	→	Final
A	n_A		$-a\xi$		$n_A - a\xi$
B	n_B		$-b\xi$		$n_B - b\xi$
C	n_C		$+c\xi$		$n_C + c\xi$
D	n_D		$+d\xi$		$n_D + d\xi$

Reaction Quotient

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$

$$\mu = \mu^\ominus + RT \ln a$$

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

where the **reaction quotient** Q is defined as

$$Q = \frac{\prod_{\text{product } j} a_j^{v_j}}{\prod_{\text{reactant } i} a_i^{v_i}}$$

Activity

$$\mu = \mu^\ominus + RT \ln a$$

- Gas: $\mu = \mu^\ominus + RT \ln(f/p^\ominus)$
- Solution
 - Solvent: $\mu_A = \mu_A^* + RT \ln a_A$
 - Solute: $\mu_B = \mu_B^* + RT \ln(K_B/p_B^*) + RT \ln a_B$
- Pure liquid and solid: ?

Approximation (perfect/ideal limit)?

Not in Atkins

Approximate Activity of Gas

$$\mu = \mu^{\ominus} + RT \ln a = \mu^{\ominus} + RT \ln(f/p^{\ominus})$$

For a perfect gas,

$$\mu = \mu^{\ominus} + RT \ln(p/p^{\ominus})$$

Hence, approximately

$$a = \frac{p}{p^{\ominus}}$$

Not in Atkins

Approximate Activity of Solute

$$\mu = \mu^\ominus + RT \ln a$$

For an ideal-dilute solution,

$$\mu_B = \mu_B^* + RT \ln(K_B/p_B^*) + RT \ln x_B$$

$$x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A} = \frac{V}{n_A} \frac{n_B}{V} = \frac{[B]}{[A]} = \frac{[B]}{c^\ominus} \frac{c^\ominus}{[A]}$$

Assuming that $[A]$ is constant,

$$\mu_B = \underbrace{\mu_B^* + RT \ln(K_B/p_B^*) - RT \ln([A]/c^\ominus)}_{\mu_B^\ominus} + RT \ln \underbrace{([B]/c^\ominus)}_a$$



Not in Atkins

Approximate Activity of Solvent

$$\mu = \mu^\ominus + RT \ln a$$

For an ideal-dilute solution,

$$\mu_A = \mu_A^* + RT \ln x_A$$

$$x_A = \frac{n_A}{n_A + n_B} \approx \frac{n_A}{n_A} = 1$$

Hence, approximately

$$a = 1$$

Pure Solids and Liquids

$$\Delta G_{\text{m}} = \mu_f - \mu_i = \int_{p_i}^{p_f} V_{\text{m}} \, dp$$

Let the initial state to be the standard state:

$$\mu - \mu^{\ominus} = \int_{p^{\ominus}}^p V_{\text{m}} \, dp' \approx V_{\text{m}} \int_{p^{\ominus}}^p 1 \, dp'$$

From the definition of an activity, $\mu - \mu^{\ominus} = RT \ln a$,

$$\ln a = \frac{V_{\text{m}}}{RT} \int_{p^{\ominus}}^p 1 \, dp' = \frac{V_{\text{m}}}{RT} (p - p^{\ominus})$$

Example: Activity of Coke

The density of coke at 1000°C is about 1.5 g cm⁻³. Calculate the activity of C(s) in the form of coke at 100 bar and 1000°C.

The molar volume is

$$V_m = (\text{molar mass})/(\text{density}) = (12 \text{ g/mol})/(1.5 \text{ g cm}^{-3}) = 8.0 \text{ cm}^3 \text{ mol}^{-1}$$

Thus,

$$\begin{aligned}\ln a &= V_m(p - p^\ominus)/RT \\ &= (8.0 \text{ cm}^3 \text{ mol}^{-1}) \times (100 - 1) \text{ bar} / (0.082 \text{ L bar K}^{-1} \text{ mol}^{-1}) / (1273 \text{ K}) \\ &= 0.0076\end{aligned}$$

or $a = 1.01$.



Pure Solids and Liquids

$$\ln a = \frac{V_m}{RT} (p - p^\ominus)$$

At moderate pressures, the activity is very close to 1.

Hence, we usually take $a = 1$ for pure solids and liquids.

Approximation for Activity

State	Measure	Approximation for a_j	Definition
Solute	molality	$b_j/b^\ominus$	$b^\ominus = 1 \text{ mol kg}^{-1}$
	molar concentration	$[J]/c^\ominus$	$c^\ominus = 1 \text{ mol dm}^{-3}$
Gas phase	partial pressure	$p_j/p^\ominus$	$p^\ominus = 1 \text{ bar}$
Pure solid, liquid		1 (exact)	

For Gaseous Reactions

$$K_p = \frac{\prod_{\text{product } j} a_j^{\nu_j}}{\prod_{\text{reactant } i} a_i^{\nu_i}} = \frac{\prod_{\text{product } j} (p_j/p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (p_i/p^\ominus)^{\nu_i}}$$

Since $p_i/p^\ominus = [i](RT/p^\ominus) = [i]/c^\ominus (c^\ominus RT/p^\ominus)$,

$$K_p = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}} \left(\frac{c^\ominus RT}{p^\ominus} \right)^{\sum_j \nu_j - \sum_i \nu_i}$$

$$K_p = K_c \left(\frac{c^\ominus RT}{p^\ominus} \right)^{\Delta \nu} = K_c \left(\frac{T}{12.03 \text{ K}} \right)^{\Delta \nu}$$

Le Chatelier's Principle

When subjected to a disturbance,
a system at equilibrium tends to respond in a way
that minimizes the effect of the disturbance.

Disturbances to consider:

- Amount
- Volume & pressure (for a gaseous reaction)
- Temperature

Amount Change

- **Increase** of the amount of a particular species
 - The system responds to **decrease** the amount
- **Decrease** of the amount of a particular species
 - The system responds to **increase** the amount
- Reactant increase / product decrease: $Q < K$
 - Decrease of reactants & increase of products
- Reactant decrease / product increase: $Q > K$
 - Increase of reactants & decrease of products

Volume/Pressure Change

For a gaseous reaction,

- Increase of the volume → decrease of the pressure
- Decrease of the volume → increase of the pressure
- Pressure change due to addition of inert gas does not affect the equilibrium

$$K_p = \frac{\prod_{\text{product } j} (p_j/p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (p_i/c^\ominus)^{\nu_i}} = \frac{\prod_{\text{product } j} (n_j RT/V p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (n_i RT/V p^\ominus)^{\nu_i}}$$

Volume/Pressure Change

$$K_p = \frac{\prod_{\text{product } j} (n_j RT / V p^\ominus)^{v_j}}{\prod_{\text{reactant } i} (n_i RT / V p^\ominus)^{v_i}}$$

For a volume change from V to xV ,

$$Q_p = \frac{\prod_{\text{product } j} (n_j RT / xV p^\ominus)^{v_j}}{\prod_{\text{reactant } i} (n_i RT / xV p^\ominus)^{v_i}}$$

$$Q_p = K_p x^{-(\sum_j v_j - \sum_i v_i)} = K_p x^{-\Delta v}$$

Volume/Pressure Change

$$Q_p = K_p x^{-\Delta\nu}$$

For $x > 1$ (volume increase),

- $\Delta\nu = 0$: $Q_p = K_p$ and the system is at equilibrium
- $\Delta\nu > 0$: $Q_p < K_p$ and the forward reaction is favored
- $\Delta\nu < 0$: $Q_p > K_p$ and the reverse reaction is favored

Temperature Change

- Exothermic reactions
 - Increased temperature favors the reactants
 - Decreased temperature favors the products
- Endothermic reactions
 - Increased temperature favors the products
 - Decreased temperature favors the reactants

van't Hoff Equation

$$\ln K = -\Delta_r G^\ominus / RT$$

Differentiating both sides with respect to T ,

$$\frac{d \ln K}{dT} = -\frac{1}{R} \frac{d(\Delta_r G^\ominus / T)}{dT}$$

Using the Gibbs-Helmholtz equation,

$$-R \frac{d \ln K}{dT} = \frac{d(\Delta_r G^\ominus / T)}{dT} = -\frac{\Delta_r H^\ominus}{T^2}$$

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

T*-Dependence of *K

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

Integration gives

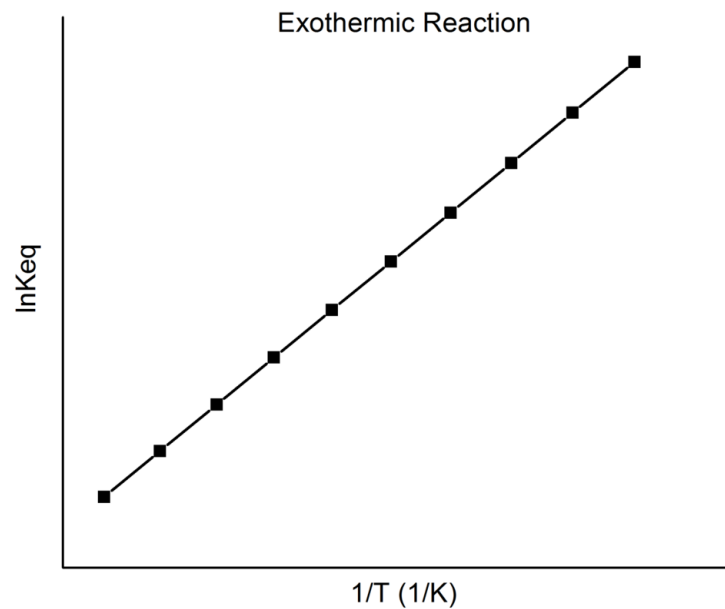
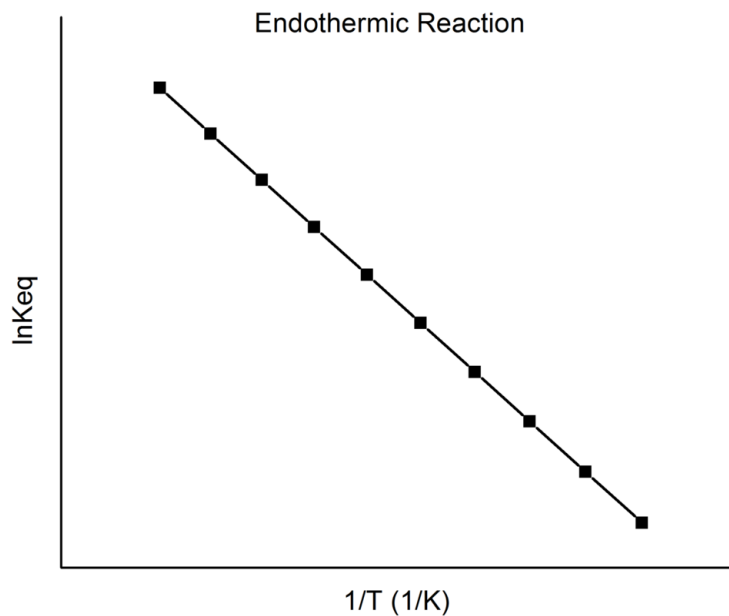
$$\int_{\ln K_1}^{\ln K_2} d \ln K = \int_{T_1}^{T_2} \frac{\Delta_r H^\ominus}{RT^2} dT$$

Assuming that $\Delta_r H^\ominus$ is constant,

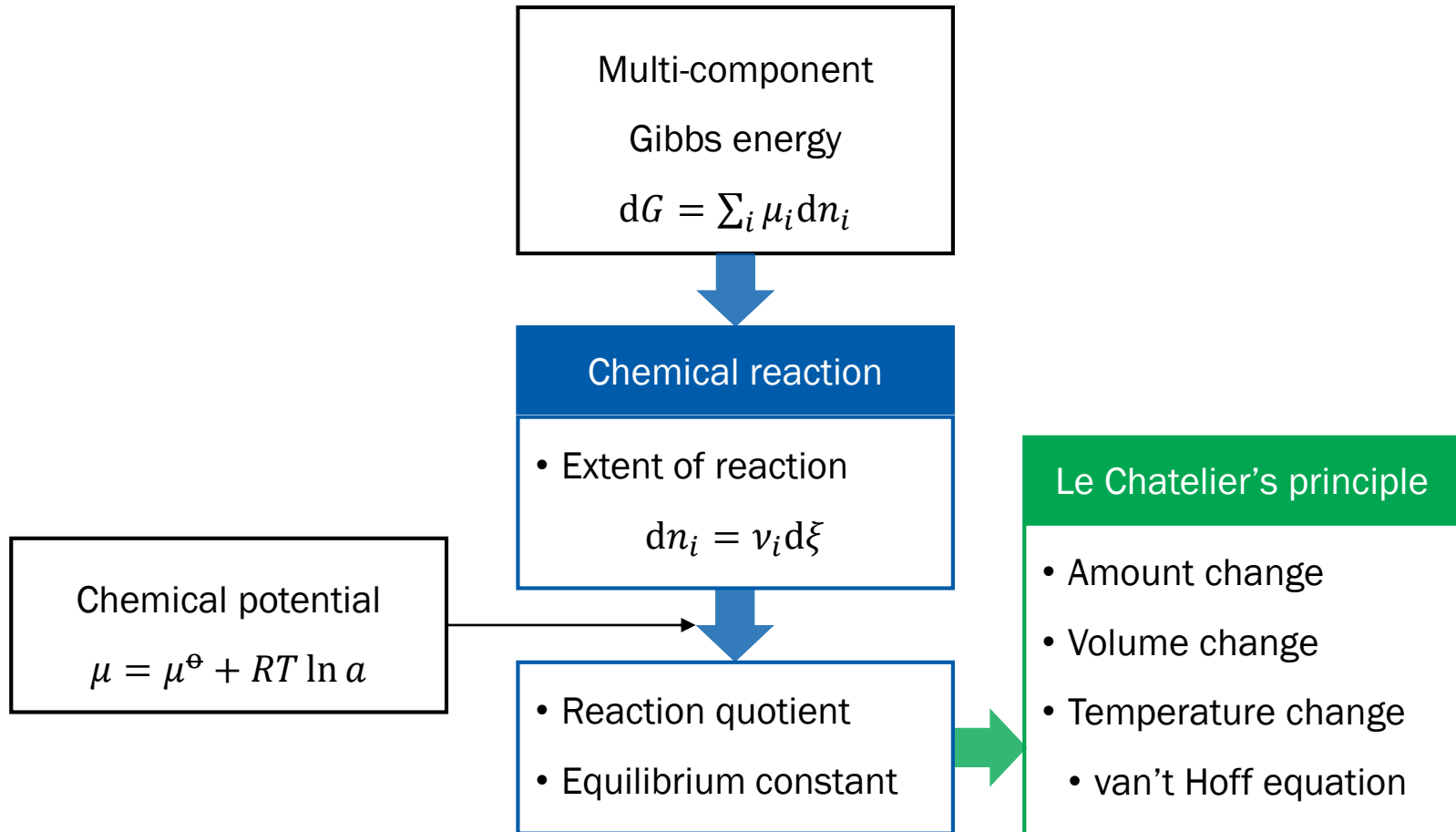
$$\ln K_2 - \ln K_1 = \frac{\Delta_r H^\ominus}{R} \int_{T_1}^{T_2} \frac{1}{T^2} dT = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

T-Dependence of *K*

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Summary



10. Electrochemistry

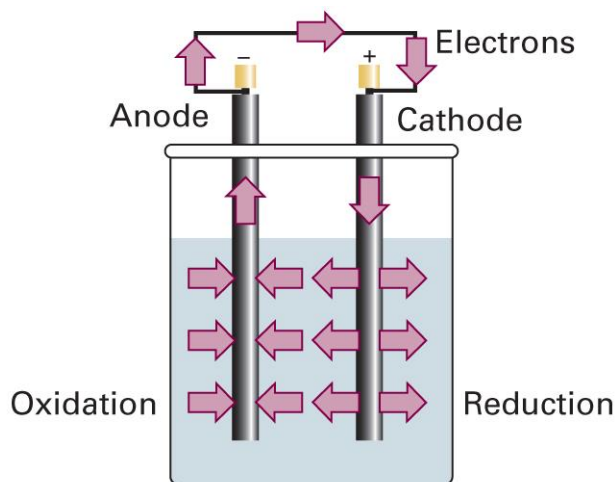
Chapter Overview

- Electrochemical cells
 - Basic terminology and concepts
- Thermodynamics of cells
 - Cell potential and reaction
 - Electrode potentials



Electrochemical Cell

- Two electrodes + electrolytes
 - Anode: oxidation
 - Cathode: reduction



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 6C.1)

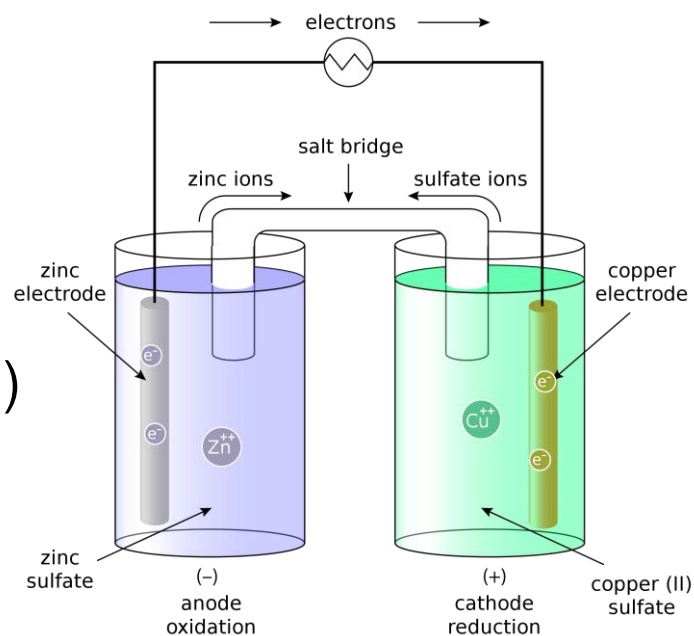
Half Cells and Cell Diagram

- Cell with $\text{Zn}^{2+}(\text{aq})/\text{Zn}(\text{s})$ and $\text{Cu}^{2+}(\text{aq})/\text{Cu}(\text{s})$ electrodes

anode: $\text{Zn}(\text{s}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-}$

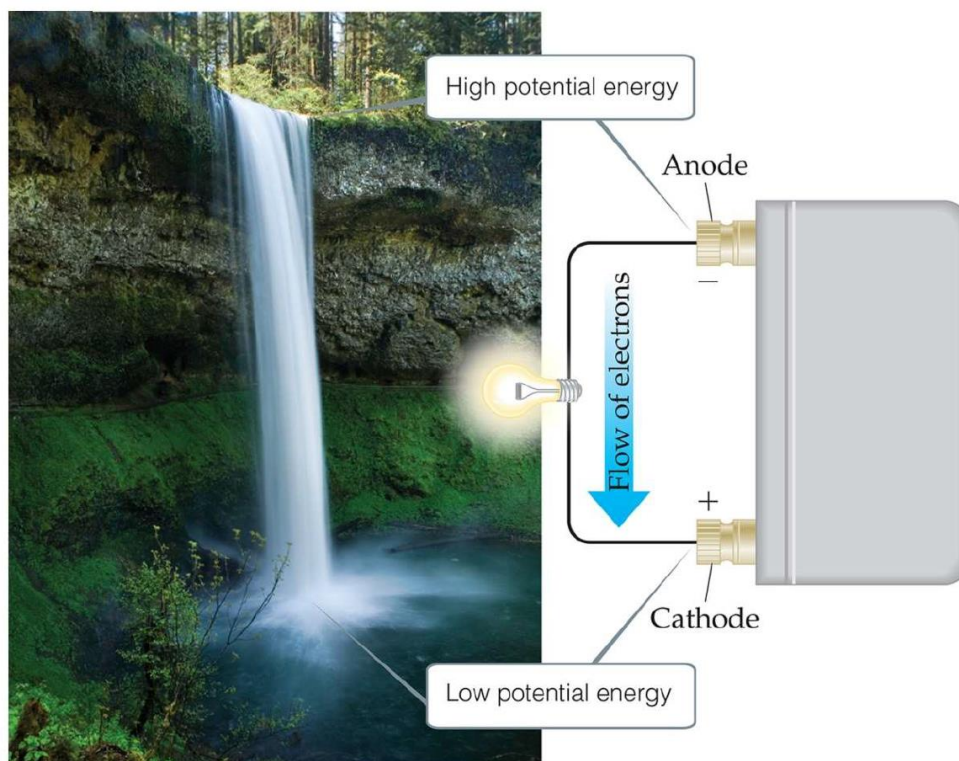
cathode: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$

- Cell diagram



Cell Potential

Potential difference between the two electrodes



(Image: <https://www.unf.edu/~michael.lufaso/chem2046/2046chapter20.pdf>)

Thermodynamics of Cell

Recall that

$$dG_{T,p} = \delta w_{\text{add,max}} = \delta w_{\text{elec}}$$

for a **reversible** electrochemical reaction.

For an infinitesimal progress of the reaction $d\xi$,

$$dG = \Delta_r G d\xi$$

Hence,

$$\delta w_{\text{elec}} = \Delta_r G d\xi$$

Electrical Work

$$\delta w_{\text{elec}} = \phi dQ$$

where ϕ is the electric field and dQ is the transferred charge.

For the electrochemical reaction,

$$\phi = E_{\text{cell}}$$

$$dQ = -\nu e N_A d\xi = -\nu F d\xi$$

Hence,

$$\delta w_{\text{elec}} = -\nu F E_{\text{cell}} d\xi$$

Thermodynamics of Cell

$$\delta w_{\text{elec}} = \Delta_r G d\xi$$

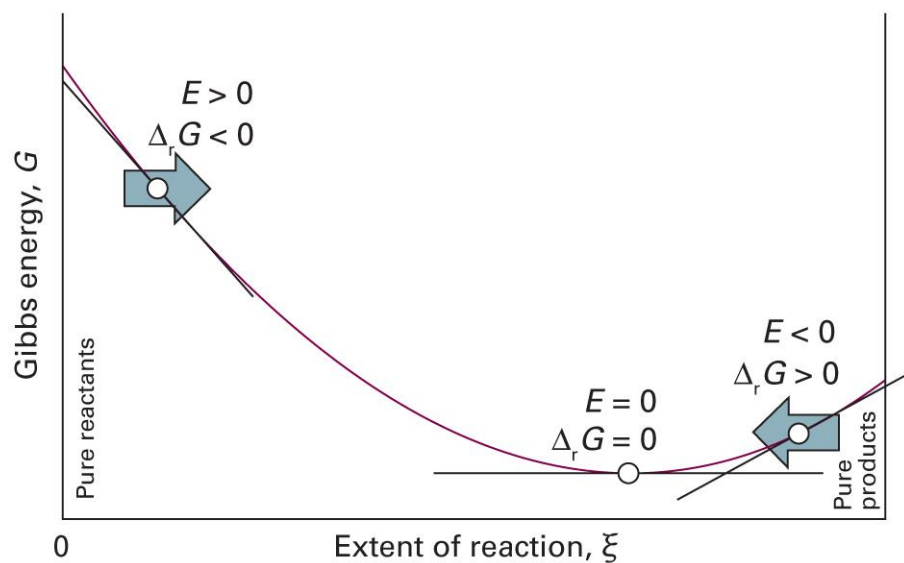
$$\delta w_{\text{elec}} = -\nu F E_{\text{cell}} d\xi$$

Combining the two, we obtain

$$\Delta_r G = -\nu F E_{\text{cell}}$$

Cell Potential and Reaction

$$\Delta_r G = -\nu F E_{\text{cell}}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 6C.4)

Nernst Equation

$$\Delta_r G = -\nu F E_{\text{cell}}$$

Using $\Delta_r G = \Delta_r G^\ominus + RT \ln Q$,

$$-\nu F E_{\text{cell}} = \Delta_r G^\ominus + RT \ln Q$$

Thus,

$$E_{\text{cell}} = E_{\text{cell}}^\ominus - \frac{RT}{\nu F} \ln Q$$

where $E_{\text{cell}}^\ominus = -\Delta_r G^\ominus / \nu F$

At Equilibrium

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln Q$$

At equilibrium, $E_{\text{cell}} = 0$ and $Q = K$

$$0 = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln K$$

or,

$$E_{\text{cell}}^{\ominus} = \frac{RT}{\nu F} \ln K$$

Determination of K_{sp}

Solubility product K_{sp} for $\text{AgCl(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq})$?

Construct an electrochemical cell with the half-reactions:

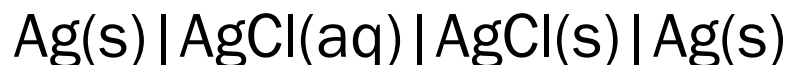
- **anode:** $\text{Ag(s)} \rightarrow \text{Ag}^+(\text{aq}) + \text{e}^-$
- **cathode:** $\text{AgCl(s)} + \text{e}^- \rightarrow \text{Ag(s)} + \text{Cl}^-(\text{aq})$

Hence, the cell diagram is



Determination of K_{sp}

Solubility product K_{sp} for $\text{AgCl(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq})$?



Measurement gives $E_{\text{cell}}^{\ominus} = -0.577 \text{ V}$. Hence,

$$E_{\text{cell}}^{\ominus} = \frac{RT}{\nu F} \ln K_{sp}$$

$$K_{sp} = e^{\nu F E_{\text{cell}}^{\ominus} / RT} = 1.77 \times 10^{-10} \text{ (at 298 K)}$$

Temperature Coefficient

$$E_{\text{cell}}^{\ominus} = -\frac{\Delta_{\text{r}}G^{\ominus}}{\nu F}$$

Since $(\partial G / \partial T)_p = -S$,

$$\frac{dE_{\text{cell}}^{\ominus}}{dT} = -\frac{1}{\nu F} \left(\frac{\partial \Delta_{\text{r}}G^{\ominus}}{\partial T} \right)_p = \frac{\Delta_{\text{r}}S^{\ominus}}{\nu F}$$

Furthermore,

$$\Delta_{\text{r}}H^{\ominus} = \Delta_{\text{r}}G^{\ominus} + T\Delta_{\text{r}}S^{\ominus} = -\nu F \left(E_{\text{cell}}^{\ominus} - T \frac{dE_{\text{cell}}^{\ominus}}{dT} \right)$$

Electrode Potentials

Cell potential = cathode contribution + anode contribution

For a cell L||R,

$$E_{\text{cell}}^{\ominus} = E^{\ominus}(\text{R}) - E^{\ominus}(\text{L})$$

Impossible to measure each contribution

Convention

- Standard hydrogen electrode (SHE): $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq})$

$E^\ominus = 0$ at all temperatures

- Standard potential $E^\ominus(\text{X})$

For a cell $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq}) || \text{X}$, $E^\ominus(\text{X}) = E_{\text{cell}}^\ominus$

Standard Potentials

Couple	$E^\ominus/\text{V}$
$\text{Ce}^{4+}(\text{aq}) + \text{e}^- \rightarrow \text{Ce}^{3+}(\text{aq})$	+1.61
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$\text{AgCl}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq})$	+0.22
$\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2(\text{g})$	0
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71

Summary

